JUMPER
solder D17 D16 if active high for error from BDUCU
solder D14 D16 if active low for error from BDUCU

J11
Conn_01x02
AMS_err

RESET_BTN
J15
Conn_01x02
+12V
R14
R_US
GND
AMS_err
optocoupler
R40
R_US
GND
falling_edge_trigger2
A
signal_in
A'
signal_out
File: falling_edge_trigger_v2.kicad_sch

D17
1N4148
AMS_err
diode OR
D14
1N4148
R37
R_US
GND
AMS_err_sig
D9
1N4148
AMS_SCS_err
D15
1N4148
error_high_latch_low2
signal_in
signal_out
reset
AMS_latch
AMS_power_on_reset
R36
R_US
GND
File: latch_to_low.kicad_sch

J13
RESET_BTN
Conn_01x04
+12V
AMS_LED+
IMD_LED+
IMD_err
R15
R_US
GND
falling_edge_trigger
A
signal_in
A'
signal_out
File: falling_edge_trigger_v2.kicad_sch

D18
1N4148
IMD_err
diode OR
D16
1N4148
R38
R_US
GND
IMD_err_sig
D8
1N4148
IMD_SCS_err
D12
1N4148
error_high_latch_low
signal_in
signal_out
reset
IMD_latch
IMD_power_on_reset
R35
R_US
GND
File: latch_to_low.kicad_sch

SDC_IN
J8
Conn_01x02

SDC_OUT
J12
Conn_01x04

RELAY

D3
1N4148
K1
SANYOU_SRD_Form_C
+12V
SDC_IN
NC
NO
R_US
R16
D5
LED
R5
10k
Q1
AO3400A
R6
100k
GND
IMD_latch

J7
Conn_01x02
D6
1N4148
K2
SANYOU_SRD_Form_C
+12V
NC
SDC_OUT
NO
R_US
R17
D7
LED
R7
10k
Q2
AO3400A
R11
100k
GND
AMS_latch

ERR LED WHEN LATCH
+12V
R34
R_US
IMD_latch
AO3407A
Q3
IMD_LED+

+12V
R33
R_US
AMS_latch
AO3407A
Q4
AMS_LED+

power cycle reset
+12V
R24
10k
C1
4.7u
D10
1N4148
IMD_power_on_reset
R22
1M
long enough to reset IMD latch while wsiting IMD input to settle down
D11
1N4148
AMS_power_on_reset

SCS for IMD and AMS
+12V
R25
1k
R26
100k
U5A
LM393
R23
10k
AMS_err
U5B
LM393
R27
100k
R28
1k
R29
1k
R30
100k
U6A
LM393
R41
10k
U6B
LM393
IMD_err
R31
100k
R32
1k
R49
R_US
AMS_SCS_err
R39
100k
Q11
AO3400A
R50
R_US
IMD_SCS_err
Q10
AO3400A
R42
100k
LOW if compliant(fall within range)
HIGH if out of range
U5C
LM393
C8
100n
U6C
LM393
C9
100n
output from lm393 directly is low for scs err and high for normal, inverted

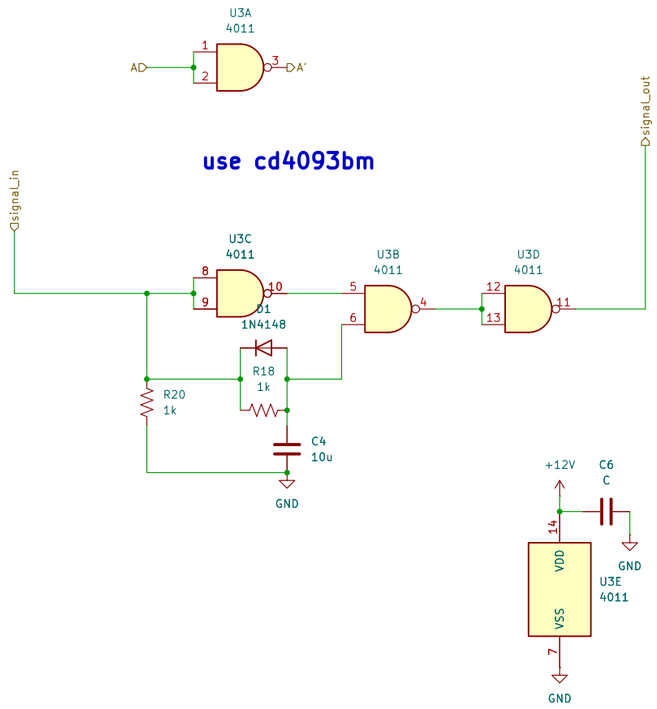
+12V
C10
1u
R12
1M
C11
1u
R46
1M
U7
TLC555xD
RST
DISCH
THRES
TRIG
CONT
VCC
OUT
GND
R13
R_US
R43
10k
R45
100k
R44
R_US
Q7
AO3400A
STARTUP_BLINK

ERR LED WHEN STARTUP
+12V
R47
R_US
AO3407A
Q9
IMD_LED+
R48
R_US
AO3407A
Q8
AMS_LED+

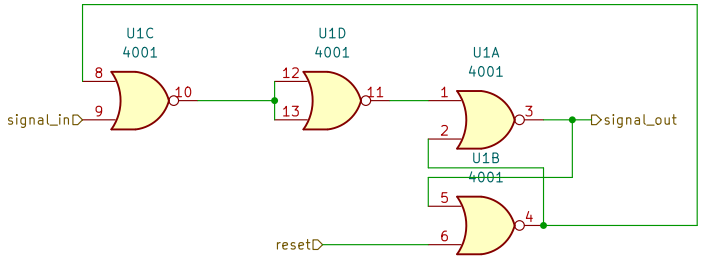
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole

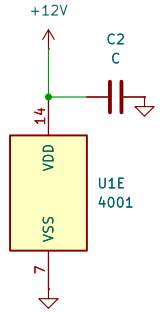
Refer to IR155-3204 datasheet
+12V
IMD_KL15_PWR
IMD_KL31_GND
J4
Conn_01x04
GND
IMD_err
IMD_OKHS
IMD OKHS low when error
R8
2.2k
R9
2.2k
IMD_MHS
J3
Conn_01x04
R10
1k
external IMD led
EXT_LED+
D4
LED
EXT_LED-

Sheet: /
File: IMD_AMS_latch.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 1/5

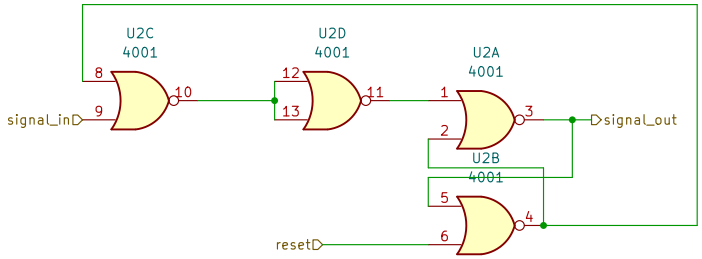


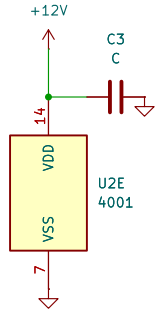
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File: falling_edge_trigger_v2.kicad_sch		
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 2/5



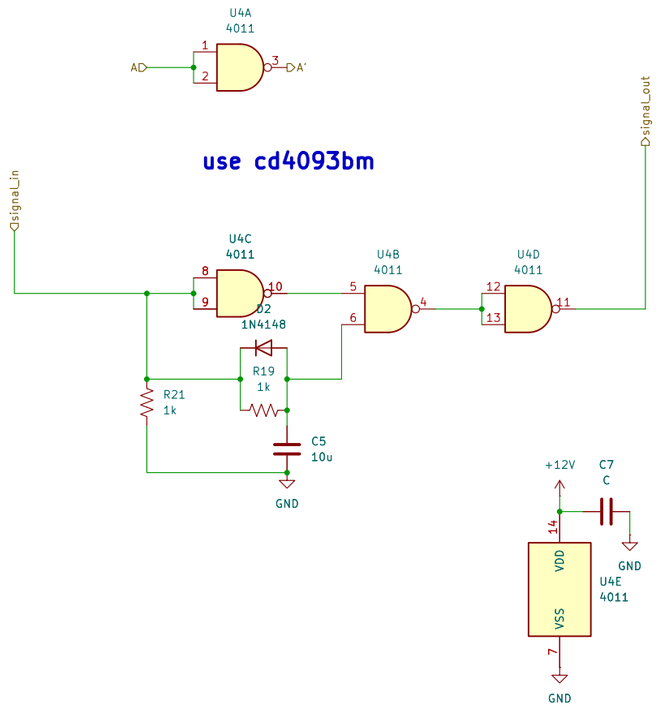


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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/5





Sheet: /error_high_latch_low2/ File: latch_to_low.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 6/5



Sheet: /falling_edge_trigger2/		
File: falling_edge_trigger_v2.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 7/5